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Research Interests

- Formal mathematics and AI, especially autoformalisation in Lean and the design of tools that translate mathematical language into machine-verifiable form.
- Low-dimensional geometry and topology, with particular interest in geometric structures on surfaces, holonomy representations, and dynamics of mapping class group actions.
- Infinite-type surfaces and related questions

Education

2019–2025 **PhD in Mathematics**

Indian Institute of Science Bangalore

Advisors: Prof. Subhojoy Gupta and Prof. Siddhartha Gadgil.

2013–2018 **BS-MS, Major in Mathematics**

Indian Institute of Science Education and Research Bhopal

MS thesis advisor: Prof. Kashyap Rajeevsarathy.

Academic Appointments and Professional Experience

Oct 2025–Present **Postdoctoral Researcher**

Department of Mathematics, Indian Institute of Science Bangalore

Working with Prof. Siddhartha Gadgil on LeanAide, an open-source project at the intersection of AI and formal mathematics. The project is supported by the AI for Mathematics Fund managed by Renaissance Philanthropy and XTX Markets.

May 2025–Present **Human Frontier Collective Fellow**

Scale AI

Designing frontier-level mathematics problems and evaluating solutions produced by state-of-the-art generative AI models.

Nov 2024–Jul 2025 **Research Intern**

ARC-Net

Worked on autoformalisation, including translating natural-language mathematics into Lean, testing LeanAide, and suggesting feature improvements.

2016–2018 **Content Development Intern**
Centre for Research in Advanced Technology for Education in Science (CREATES)
Served during Summer 2016, Summer 2017, and Sep–Nov 2018; developed undergraduate-level mathematics quizzes and educational content.

Teaching and Mentoring

Apr–May 2026 **Lead Coordinator and Tutor**
Hackathon: LeanLang for Verified Autonomy, IISc Bangalore + Online
Serving as lead coordinator and tutor for the hackathon, organised in association with Emergence India Labs.

Dec 2025 **Tutor, LeanLang for Programmers**
Indian Institute of Science Bangalore
Tutor for an informal workshop introducing Lean in mathematical and programming contexts.

Jun 2023 **Tutor, Lie Groups**
Advanced Instructional School in Lie Groups and Lie Algebras
Helped students with problem sheets and discussions during the school.

Dec 2021 **Tutor, Real Analysis**
Winter School for Young Women in Mathematics
Held problem-solving sessions and answered questions on course material.

Sem. I 2022–23 **Teaching Assistant, UM 201**
Department of Mathematics, IISc Bangalore
Probability and Statistics. Conducted tutorials and assisted with grading examinations and quizzes.

Sem. I 2021–22 **Teaching Assistant, MA 200**
Department of Mathematics, IISc Bangalore
Multivariable Calculus. Conducted tutorials and assisted with grading assignments and examinations.

Sem. I 2020–21 **Teaching Assistant, MA 261**
Department of Mathematics, IISc Bangalore
Probability Models. Conducted tutorials and assisted with grading examinations and quizzes.

Sem. I 2017–18 **Teaching Assistant, HSS 101**
Department of Humanities and Social Sciences, IISER Bhopal
English for Communication. Assisted with grading.

Fellowships

- 2019–2024 **NBHM PhD Fellowship**, Department of Atomic Energy, India.
2013–2018 **INSPIRE Fellowship**, Department of Science and Technology, India.
Summer 2015 **Summer Research Fellow**, Indian Academy of Sciences.

Publications

- 2026 **From discrete to dense: explorations in the moduli space of triangles**, with Aahana Aggarwal and Subhojoy Gupta. Accepted for publication in *Mathematics Magazine*.
2025 **On dynamics of the Mapping class group action on relative $\mathrm{PSL}(2, \mathbb{R})$ -character varieties**. *Groups, Geometry and Dynamics*.
2024 **The Goldman bracket characterizes homeomorphisms between non-compact surfaces**, with Siddhartha Gadgil and Sumanta Das. *Algebraic and Geometric Topology*.
2023 **Factoring periodic maps into Dehn twists**, with K. Rajeevsarathy and N. K. Dhanwani. *Journal of Pure and Applied Algebra*, 227(1).

Talks

- Nov 2025 **Mapping class group dynamics on $\mathrm{PSL}(2, \mathbb{C})$ -character varieties**, In House Student-PostDoc Symposium, IISc Bangalore.
Sep 2024 **The Goldman bracket characterises the homeomorphisms between non-compact surfaces**, World of GroupCraft IV. *[Video]*.
Apr 2024 **When is a homotopy equivalence of a surface homotopic to a homeomorphism?**, In House Student-PostDoc Symposium, IISc Bangalore.
Jan 2023 **On Three Theorems of Surface Homeomorphisms**, Prime Time Seminars, IISc Bangalore.
Nov 2022 **On Three Theorems of Surface Homeomorphisms**, IISER Bhopal Geometry and Topology Seminars. *[Video]*.
Nov 2021 **Conformal Bijections, Biholomorphisms and Hyperbolic Isometries**, Prime Time Seminars, IISc Bangalore.

Academic Service

- 2022–2025 Co-organiser, Students' Topology Geometry Seminars, Department of Mathematics, IISc.
2023 Co-Organiser, Postdoc-Student Symposium, Department of Mathematics, IISc.

- 2021–2023 Co-organiser, Prime Time Seminars, Department of Mathematics, IISc.
- 2017 Student Volunteer, Regional Workshop on Research and Opportunities, Indian Women in Mathematics (IWM).
- Nov 2016 Student Volunteer, Indian Academy of Sciences 82nd Annual Meeting, IISER Bhopal.

Conferences and Workshops

- Apr–May 2026 Hackathon: LeanLang for Verified Autonomy, IISc Bangalore + Online, in association with Emergence India Labs.
- Dec 2025 LeanLang for Programmers, IISc Bangalore.
- Apr 2025 Lean for the Curious Mathematician, ICTS Bengaluru.
- Feb 2025 New Trends in Teichmüller Theory, ICTS Bengaluru.
- Jan 2025 Lean Together 2025, Online.
- Jul–Aug 2024 Geometry in Groups, ICTS Bengaluru.
- Sep 2023 Low-Dimensional Topology, IISER Pune.
- May 2023 1,2,3: Curves, Surfaces and 3-manifolds, Technion - Israel Institute of Technology, Haifa, Israel.
- Dec 2022 Ergodic Theory and Dynamical Systems, ICTS Bengaluru.
- Jul 2019 Advanced Instructional School in Riemannian Geometry, NCM at IISc Bangalore.
- Oct 2017 Regional Workshop on Research and Opportunities, Indian Women in Mathematics (IWM), IISER Bhopal.
- Jan 2015 Indian Science Congress, University of Mumbai.

Leadership and Community Involvement

- 2016–2017 Member, Central Advisory Committee of the Student Activity Council, IISER Bhopal.
- Apr 2016 Member, Organising Committee, Luminaire '16 movie festival, IISER Bhopal.
- Oct 2015 Organiser, Arts and Letters '15 literary festival, IISER Bhopal.
- Jan 2015 Member, Core Organising Committee, Singularity '15 science festival, IISER Bhopal.
- 2014–2015 Member, Prayaas social initiative club, IISER Bhopal.
- 2014–2015 Vice-Secretary, Fine Arts and Literary Council, Student Activity Council, IISER Bhopal.
- 2014–2015 Coordinator, Movie Club, IISER Bhopal.

Technical Skills

Programming	Python, Mathematica, Haskell (beginner), Lean (beginner).
Libraries and Tools	pandas, NumPy, scikit-learn, matplotlib.

Languages

English	Fluent in speaking, reading, and writing.
Telugu	Fluent in speaking, reading, and writing.
Hindi	Fluent in speaking, reading, and writing.
Malayalam	Fluent in speaking; intermediate in reading and writing.

References

Prof. Subhojoy Gupta

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Prof. Siddhartha Gadgil

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Prof. Kashyap Rajeevsarathy

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